

Apache Fineract Setup Guide (Step-by-Step)

This guide provides a clear, step-by-step process for setting up Apache Fineract 1.x locally using Java 17, Maven, MySQL, and Spring Boot. It's designed for interns or developers who are starting to work with the Apache Fineract platform for microfinance and loan management systems.

1■■■ Prerequisites

- Java JDK 17 or above - Apache Maven 3.8+ - MySQL Server 8.x - Git (latest version) - IDE: IntelliJ IDEA / Eclipse / VS Code - Postman (optional for API testing)

2■■■ Clone the Source Code

```
git clone https://github.com/apache/fineract.git cd fineract
```

3■■■ Set Up MySQL Database

```
mysql -u root -p CREATE DATABASE fineract_tenants; CREATE DATABASE mifostenant_default;  
CREATE USER 'fineract'@'localhost' IDENTIFIED BY 'password'; GRANT ALL PRIVILEGES ON  
*. * TO 'fineract'@'localhost'; FLUSH PRIVILEGES;
```

4■■■ Configure Application Properties

```
Edit the file: fineract-provider/src/main/resources/application.properties # Database settings  
fineract_tenants.username=fineract fineract_tenants.password=password  
fineract_tenants.host=localhost fineract_tenants.port=3306 fineract_default.username=fineract  
fineract_default.password=password fineract_default.host=localhost fineract_default.port=3306
```

5■■■ Build the Project

```
mvn clean install -DskipTests (If build fails: mvn clean install -DskipTests -Dmaven.test.skip=true)
```

6■■■ Run Apache Fineract

```
cd fineract-provider mvn spring-boot:run Access the API:  
http://localhost:8443/fineract-provider/api/v1/
```

7■■■ Default Login

Username: mifos Password: password

8■■■ Test Using Postman

```
GET http://localhost:8443/fineract-provider/api/v1/clients Header: Authorization: Basic  
bWlmb3M6cGFzc3dvcmQ=
```

9■■■ Optional: Run with Docker

```
git clone https://github.com/apache/fineract.git cd fineract docker-compose up
```

■ Front-End Setup (Optional)

```
git clone https://github.com/openMF/community-app.git cd community-app npm install npm start
```

Access UI: <http://localhost:9000>

■ Setup Complete

You now have a fully running Apache Fineract instance connected to MySQL with REST API access. This setup is ideal for testing, learning, and contributing to microfinance or loan management modules.